Light at the Edge of Darkness

A numerical study of how a black hole bends and captures light

Computational Physics project write-up

1 What is this project about? (plain language)

A black hole is a region of space where gravity is so strong that nothing, not even light, can come back out once it gets inside. The boundary of that region is called the **event horizon**. Light that passes *near* a black hole but not inside it is not captured, but its path is bent. This bending is called **gravitational lensing**.

We study one clean question:

> *A ray of light comes in from far away, travelling in a straight line that would miss the black hole by a distance b. For which values of b does the ray escape, for which is it captured, and by how much is it bent when it escapes?*

The distance b is called the **impact parameter**. It is the perpendicular distance between the incoming straight line and the centre of the black hole. We measure lengths in units of $M = GM_{\rm BH}/c^2$, which is a length (about 1.5 km for a black hole of one solar mass). In these units $G = c = 1$.

Model used. A non-rotating, uncharged (Schwarzschild) black hole in empty space. Light is treated as a ray moving along a path called a null geodesic in this fixed spacetime. There is no gas, no accretion disc and no spin. We do *not* solve Einstein's field equations; we only follow light in the known solution.

Why this is a computational problem. The ray equation is nonlinear and has no solution in elementary functions. The exact bending angle can be written as an integral, but that integral does not give a simple formula for the paths, for the capture boundary or for what happens when b is close to the capture value. A computer lets us trace the rays, find the capture boundary, and measure the errors of the method. Section 3 explains this in detail.

2 The physics, step by step

2.1 The spacetime

In the equatorial plane the Schwarzschild metric is

$$ds^2 = -\Big(1 - \frac{2M}{r}\Big)dt^2 + \Big(1 - \frac{2M}{r}\Big)^{-1} dr^2 + r^2\, d\phi^2. \tag{1}$$

Here r is the distance from the centre, ϕ is the angle around it and t is time. The event horizon is at $r_h = 2M$.

2.2 Conserved quantities and the orbit equation

Light moves along paths with $ds^2 = 0$. Because the metric does not depend on t or ϕ, two quantities are conserved along the ray: the energy $E = (1 - 2M/r)\, dt/d\lambda$ and the angular momentum $L = r^2\, d\phi/d\lambda$ (λ labels points along the ray). Their ratio defines the impact parameter,

$$b = \frac{L}{E}. \tag{2}$$

Putting $ds^2 = 0$ and these constants together gives

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2 - \left(1 - \frac{2M}{r}\right)\frac{L^2}{r^2}. \tag{3}$$

Use $u = 1/r$ and divide by L^2, using $du/d\phi = -\dot{r}/L$:

$$\boxed{\left(\frac{du}{d\phi}\right)^2 = \frac{1}{b^2} - u^2 + 2Mu^3} \qquad \Rightarrow \qquad \boxed{\frac{d^2u}{d\phi^2} + u = 3Mu^2}. \tag{4}$$

The second form comes from differentiating the first. Without the $3Mu^2$ term this is $u'' + u = 0$, whose solution is a straight line. So $3Mu^2$ is the term that bends light. The first form is a *conservation law*; we use it to start each ray and to check the numerics. The second form is what the computer integrates, because it is a plain second-order ODE.

2.3 Capture, escape and the critical impact parameter

Rewrite the radial equation as $(dr/d\lambda)^2/L^2 = 1/b^2 - V(r)$ with

$$V(r) = \frac{1 - 2M/r}{r^2}. \tag{5}$$

This acts like a potential barrier for light (Fig. 1). Setting $dV/dr = 0$ gives the top of the barrier at $r_{ph} = 3M$, called the **photon sphere**, where $V = 1/(27M^2)$. A ray whose $1/b^2$ is below this peak is turned back at the barrier and escapes. A ray whose $1/b^2$ is above the peak goes over it and falls in. The dividing value is

$$\boxed{b_c = 3\sqrt{3}\,M \approx 5.196\,M}. \tag{6}$$

The three special radii are different things: $2M$ is the point of no return (horizon), $3M$ is where light could circle in an unstable orbit (photon sphere), and b_c is a *distance of the incoming ray from the axis*, not a radius at which anything sits.

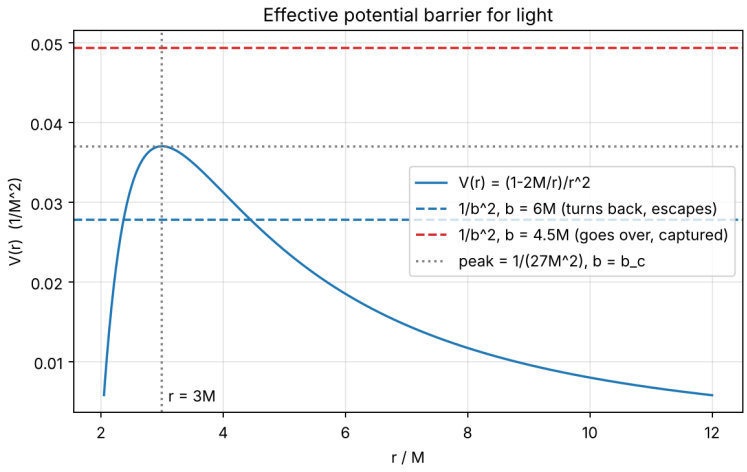


Figure 1: Effective potential $V(r)$ for light. A ray with $b = 6M$ (blue line) meets the barrier and turns back. A ray with $b = 4.5M$ (red line) passes over the top and is captured.

2.4 Deflection angle

For an escaping ray the bending angle is the total angle swept minus π (a straight line sweeps exactly π):

$$\alpha = \Delta\phi - \pi. \tag{7}$$

Weak field ($b \gg M$). Treat $3Mu^2$ as a small correction. The straight-line solution is $u_0 = \sin\phi/b$. Putting it into the right-hand side, a particular solution is $u_1 = M(1+\cos^2\phi)/b^2$. Then $u = u_0+u_1$ reaches $u = 0$ at $\phi = -2M/b$ and $\phi = \pi + 2M/b$, so

$$\alpha \approx \frac{4M}{b}. \tag{8}$$

This is the standard first-order result. It is only a first term: it fails when b is comparable to b_c.

Exact integral. Solving the first form of Eq. (4) for $d\phi$ gives

$$\alpha = 2\int_0^{u_m} \frac{du}{\sqrt{1/b^2 - u^2 + 2Mu^3}} - \pi, \tag{9}$$

where $u_m = 1/r_{\min}$ is the smallest positive root of the expression under the square root (the point of closest approach). This is an exact result, but it is an integral that has to be evaluated numerically (or with elliptic functions). We use it as a **benchmark** to test the ray tracer.

Known expansions that we test numerically. For large b: $\alpha = 4M/b + \frac{15\pi}{4}(M/b)^2 + \frac{128}{3}(M/b)^3 + \dots$ For b just above b_c (strong field), Bozza [1] gives

$$\alpha \approx -\ln\Big(\frac{b}{b_c} - 1\Big) + \ln\big[216(7 - 4\sqrt{3})\big] - \pi, \tag{10}$$

so the bending grows without limit, like a logarithm, as $b \to b_c$, and the ray goes round the black hole many times.

3 Why computation is needed

- Equation (4) is **nonlinear** (u^2 term). It does not have a solution in elementary functions. Exact results exist only as integrals or elliptic functions, which themselves need numerical evaluation.
- The weak-field formula $4M/b$ is an approximation and fails badly in the strong-field region (Table 2).
- To **see** the paths, and to find out which rays are captured and which escape for any b, we must follow each ray step by step.
- Near b_c the ray is extremely sensitive: a tiny change in b changes how many times it circles. We need numerical event detection and a careful search for b_c.
- Numerical answers have errors. A computational study has to measure them (step size, finite start and end radii, rounding) and compare with an analytic benchmark. That is the scientific content of this project.

4 Computational methods

4.1 Fourth-order Runge–Kutta (RK4)

Define $v = du/d\phi$. Then Eq. (4) becomes two first-order equations

$$\frac{du}{d\phi} = v, \qquad \frac{dv}{d\phi} = 3Mu^2 - u. \tag{11}$$

For a step h in angle, from (u, v) the RK4 method computes four slopes $k_1, \dots, k_4$ (at the start, twice at the midpoint, and at the end) and advances

$$y_{n+1} = y_n + \frac{h}{6}\,(k_1 + 2k_2 + 2k_3 + k_4), \qquad y = (u, v). \tag{12}$$

The local error per step is $O(h^5)$ and the global error is $O(h^4)$: halving h should cut the error by about 16. We wrote RK4 by hand (no library ODE solver) and also wrote the simple Euler method, $y_{n+1} = y_n + hk_1$ (global error $O(h)$), for comparison.

4.2 Starting the ray correctly

Each ray starts at a large radius r_0 with $u_0 = 1/r_0$ and $\phi_0 = \arcsin(b/r_0)$ (the position of a straight line at height b). The starting slope is *not* chosen freely: it is taken from the conservation law, $v_0 = +\sqrt{1/b^2 - u_0^2 + 2Mu_0^3}$. The positive sign means the ray is moving inward (u increasing as ϕ increases). This makes the ray carry exactly the intended b.

4.3 Deciding the fate of a ray (event detection)

- **Captured:** $u \geq 1/(2M)$, i.e. the ray reaches $r = 2M$.
- **Escaped:** $v < 0$ (moving outward) and $u \leq 1/r_{\max}$, i.e. it has reached a large outer radius $r_{\max}$.
- **Unresolved:** neither happens before the total angle reaches a set limit (40π). Such a ray is *not* called captured or escaped, and no deflection is reported for it.

Because RK4 moves in jumps of size h, the ray usually overshoots the boundary. The code then finds the exact crossing inside the last step by **bisection**. This matters: without it the order of accuracy is lost (Section 5.4).

A finite computation cannot prove that a ray stays near the photon sphere forever. It can only say what the ray did in the allowed range.

4.4 Finding b_c by bisection

We start with $b = 4M$ (captured) and $b = 7M$ (escaped). We repeatedly take the midpoint, trace that ray, and keep the half that still contains the change from captured to escaped. After 60 halvings the bracket is about 10^{-15} wide. The result is compared with $3\sqrt{3}M$.

4.5 Measuring the deflection

The ray is started at r_0 and stopped at $r_{\max}$, not at infinity. A straight line between the same two radii would sweep $\pi - \arcsin(b/r_0) - \arcsin(b/r_{\max})$. The measured bending is therefore

$$\alpha = \phi_{\text{end}} - \pi + \arcsin(b/r_{\max}), \tag{13}$$

(using $\phi_0 = \arcsin(b/r_0)$). We check how much error the finite r_0, $r_{\max}$ still leave behind.

4.6 Benchmark

The exact integral above is evaluated with **Simpson's rule**, after the substitution $u = u_m(1 - s^2)$ that removes the singularity at the turning point. The turning point u_m is found by bisection. Changing the number of Simpson points from 1000 to 4000 changes the result at $b = 8M$ by only 3×10^{-12}, so the benchmark is accurate to about 10^{-11}. Differences between numerical and exact values smaller than this are not meaningful.

5 Results

All numbers below were produced by the supplied program `bh_lensing.py` (full output in the file run_output.txt). Default settings: RK4, $h = 0.002$ rad, $r_0 = 1000M$, $r_{\max} = 2000M$.

5.1 Capture versus escape

Fig. 2 shows traced rays. Rays with $b \leq 5.1961M$ were captured and rays with $b \geq 5.1962M$ escaped, so the switch happens at $b_c = 5.19615M$. Rays just above b_c circle the black hole near $r = 3M$ before leaving.

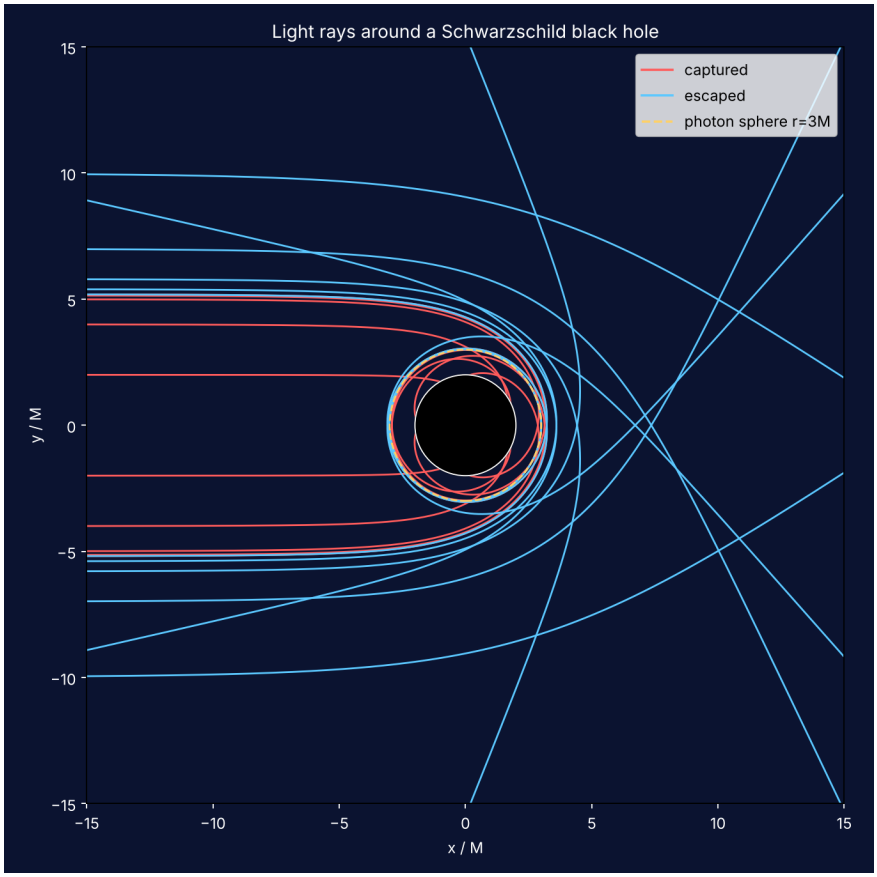


Figure 2: Traced rays for $b/M = 2$, 4, 5, 5.15, $b_c(1 \mp 10^{-6})$, 5.4, 5.8, 7, 10 (rays come in from the left; the picture is mirrored about the x-axis). Red: captured. Blue: escaped.

5.2 Critical impact parameter by bisection

Method	step h (rad)	b_c found (M)	error $b_c - 3\sqrt{3}M$
RK4	0.1	5.196151186	-1.2×10^{-6}
RK4	0.05	5.196152351	-7.2×10^{-8}
RK4	0.02	5.196152421	-1.7×10^{-9}
RK4	0.01	5.196152423	-1.1×10^{-10}
RK4	0.005	5.196152423	-6.6×10^{-12}
RK4	0.002	5.196152422706	-1.7×10^{-13}
Euler	0.001	5.197678	$+1.5 \times 10^{-3}$
Euler	0.0005	5.196915	$+7.6 \times 10^{-4}$

Table 1: Estimate of b_c (exact value $3\sqrt{3}M = 5.196152422707$). RK4 error falls by about 16 when h is halved; Euler error only halves.

The result does not depend on the start and end radii: for four settings, from $r_0 = 50M$ (with $r_{\max} = 100M$) up to $r_0 = 5000M$ (with $r_{\max} = 10000M$) the error stayed between -7×10^{-14} and -1.7×10^{-13}. Physically this is expected, because capture depends only on the barrier, and the start condition uses the exact conservation law.

5.3 Deflection angle against the exact value and the weak-field formula

b/M	RK4 α (rad)	exact α (rad)	|RK4−exact|	$(4M/b - \alpha)/\alpha$
5.3	3.557938042	3.557938042	4×10^{-11}	−79%
6	1.719388310	1.719388310	6×10^{-11}	−61%
8	0.858730018	0.858730018	1.4×10^{-10}	−42%
10	0.590395788	0.590395788	2.7×10^{-10}	−32%
20	0.236135998	0.236135995	2.1×10^{-9}	−15%
30	0.148248094	0.148248087	7.2×10^{-9}	−10%

Table 2: The ray tracer agrees with the exact integral to better than 10^{-8} rad. The weak-field formula $4M/b$ underestimates the bending by about 10% even at $b = 30M$ and by almost 80% near b_c.

At $b = 30M$ the three-term series gives 0.14800, against the exact 0.14825; the remaining gap shrinks roughly like b^{-4} (1.4×10^{-3} at $b = 20M$, 2.5×10^{-4} at $b = 30M$), as expected for a series cut off after the b^{-3} term.

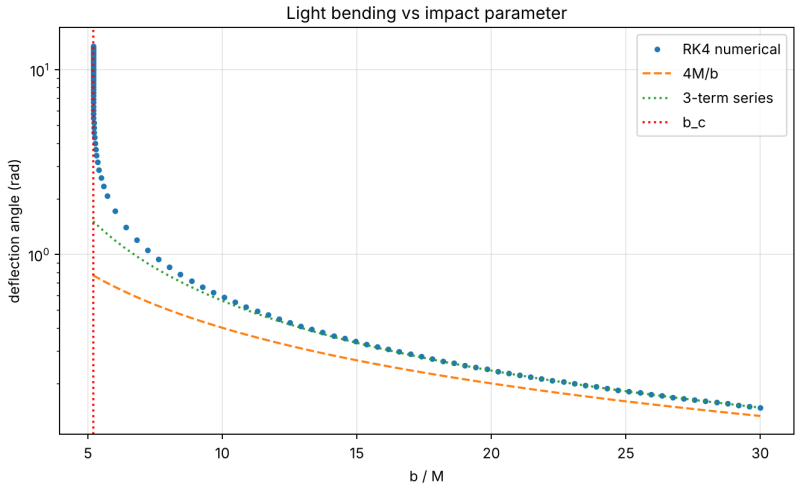


Figure 3: Deflection angle against impact parameter (log vertical axis). The first-order $4M/b$ approaches the numerical result only for large b.

5.4 Convergence with step size (and Euler against RK4)

The error is measured against a run with $h = 0.0005$; the gap of that run from the exact value is below 1.4×10^{-10}.

Case	fitted slope, $b = 8M$	fitted slope, $b = 6M$
RK4 with bisection on the last step	3.9	4.2
Euler	1.0	1.0
RK4 *without* locating the boundary exactly	0.0	0.9

Table 3: Slope of $\log|\text{error}|$ against $\log h$. A slope of 4 means fourth-order convergence.

The last row is the important warning: if the boundary crossing is not found exactly, the error is stuck at the first-order level (or does not fall at all) even though RK4 is used inside the step.

A cost comparison at $b = 8M$: RK4 with $h = 0.1$ (about 125 evaluations of the slope function) gives an error of 4.0×10^{-7}. Euler needs $h = 2.5 \times 10^{-4}$ (about 12 600 evaluations) and still has an error of 2.3×10^{-4}. RK4 is about 500 times more accurate with about 100 times less work.

The first integral (4) should stay constant along the ray. Its drift for $h = 0.1, 0.05, 0.025, 0.0125$ is 6.8×10^{-7}, 4.0×10^{-8}, 2.4×10^{-9}, 1.5×10^{-10}, again falling by about 16 per halving. This is an independent check that does not need any exact answer.

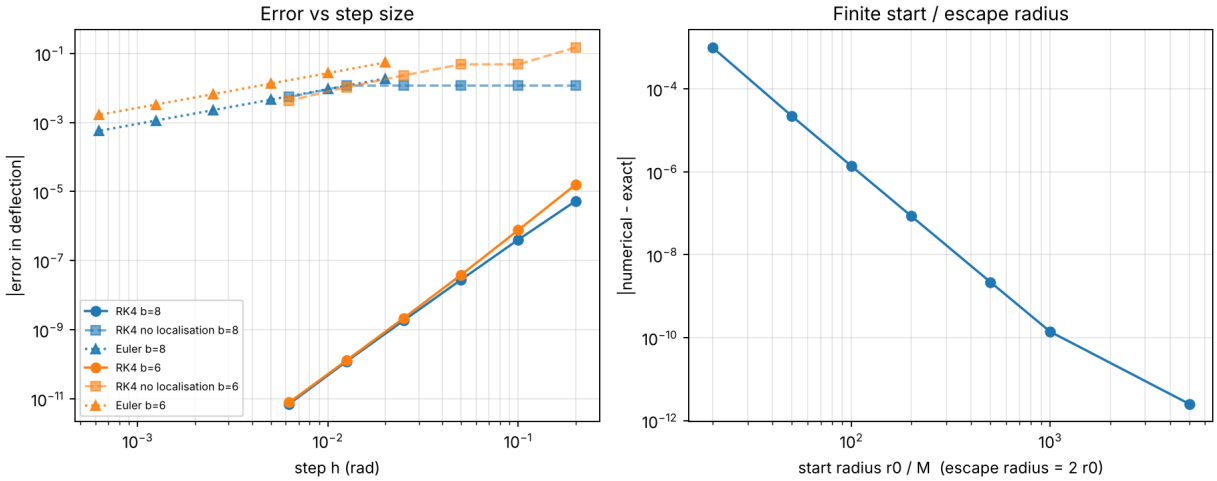


Figure 4: Left: error against step size for RK4, RK4 without exact boundary location, and Euler. Right: error caused by finite start/escape radius at $b = 8M$ (escape radius $= 2r_0$).

5.5 Finite start and escape radius

The error from starting at r_0 instead of infinity falls like r_0^{-4} (fitted slope -4.00 for $r_0 \geq 50M$): 2.2×10^{-5} at $r_0 = 50M$, 8.5×10^{-8} at $200M$, 1.4×10^{-10} at $1000M$. It grows with b: at $r_0 = 200M$ the error is 8.5×10^{-8} for $b = 8M$ and 2.2×10^{-5} for $b = 50M$. Our default $r_0 = 1000M$ keeps it below 10^{-8} for all $b \leq 30M$.

5.6 Near the critical impact parameter

$b/b_c - 1$	α (rad)	α/π
10^{-1}	2.081	0.66
10^{-2}	4.230	1.35
10^{-3}	6.511	2.07
10^{-4}	8.810	2.80
10^{-6}	13.415	4.27
10^{-8}	18.020	5.74
10^{-10}	22.626	7.20

Table 4: Deflection grows by about 2.30 rad ($= \ln 10$) for every factor of 10 closer to b_c.

A straight-line fit of α against $\ln(b/b_c - 1)$ for $10^{-8} \leq b/b_c - 1 \leq 10^{-3}$ gives slope -0.9998 (theory -1) and intercept -0.3970 (theory -0.4002). This agrees with the logarithmic law of Bozza [1]; the small difference in the intercept is expected, because the law is only the leading term.

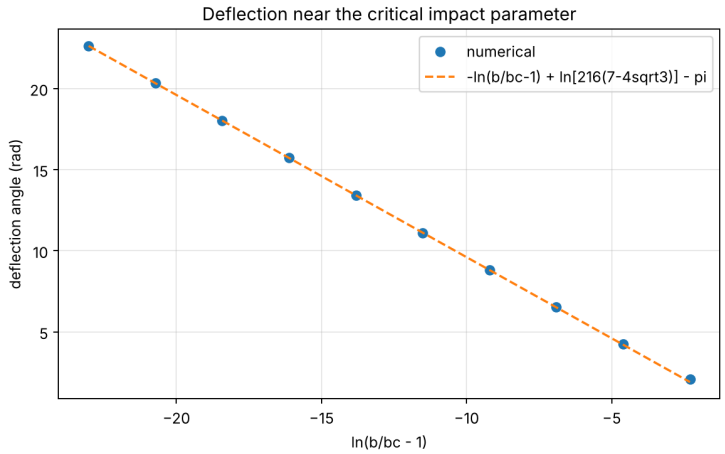


Figure 5: Numerical deflection against $\ln(b/b_c - 1)$ with the strong-field logarithmic law.

How reliable is the capture/escape decision close to b_c? The numerical method has its own tiny error, which shifts the numerical b_c (Table 1). A ray with $b = b_c(1 - 10^{-9})$ is truly captured, but with $h = 0.05$ the program wrongly reports it escaped (the numerical threshold is 7×10^{-8} too low); with $h = 0.01$ and 0.002 it is correct. At $b = b_c(1 - 10^{-11})$ only $h = 0.002$ is correct. So *the classification is trustworthy only for $|b - b_c|$ larger than the numerical shift of b_c for that step size.*

The angle limit also matters: for $b = b_c(1 + 10^{-6})$ the ray must sweep about 16.6 rad (5.3π). With limits of $2\pi, \ldots, 5\pi$ the program returns "unresolved" and with 6π it returns "escaped". This shows that "unresolved" is a real outcome, not a hidden label.

5.7 Other checks

- With the mass set to zero the code must give no bending: result 2.9×10^{-13} rad.
- Running the same ray twice gives identical numbers (the program has no randomness).

6 How the project meets the five requirements

Requirement	Where it is covered
1. A physics problem that needs computation	Capture and bending of light by a Schwarzschild black hole (Sections 1, 2)
2. Explain the physics and why computation is needed	Sections 2 and 3: nonlinear orbit equation, only integral-form exact results, need for tracing and for error measurement
3. Develop and implement a method	Section 4: hand-written RK4 (and Euler), event detection, bisection for b_c, Simpson benchmark
4. Present and analyse the results	Section 5: figures and tables, physical interpretation, strengths and limits of the method
5. At least one extension or parameter study	Section 5: step size, Euler against RK4, finite start/escape radius, behaviour near b_c, angle limit

7 Course topics used

- Ordinary differential equation solvers, Euler and Runge–Kutta (the ODE lecture notes).
- Error order, step size and stability (the notes on the Euler method and stability).

- Bisection (the root-finding notes; the shooting-method notes also combine bisection with RK4, as here).
- Numerical integration with Simpson's rule (the integration notes).

Only methods that are actually used are listed.

8 Limitations

- The black hole is non-rotating and uncharged, and space around it is empty. Real black holes spin and have gas around them.
- We follow rays in a fixed known spacetime; we do not solve Einstein's equations.
- We trace light rays only. We do not make a telescope image of a real black hole. The observed ring-shaped shadow in real data (for example, [4]) is related to b_c, but producing such an image needs much more work.
- Rays start at a finite radius, not at infinity. The resulting error is measured above and is small for our settings.
- Close to b_c the answer depends on step size (see above). A finite calculation cannot prove that a ray would stay near $r = 3M$ forever.
- The physics here is standard and well known. The contribution of the project is the implementation, the validation and the error analysis, not new physics.

Possible extension: trace rays backward from an observer's screen to construct a simplified black-hole shadow. This would need new code and new checks.

9 Guide to the program

Function	Job
`rk4`, `euler`, `step`	one step of the chosen method for (u, v)
`trace`	follows one ray; returns fate (0 unresolved, 1 captured, 2 escaped), final angle, first-integral drift, path
`defl`	deflection angle of an escaping ray (NaN otherwise)
`find_bc`	bisection on b to find the capture boundary
`umax`, `exact`	exact deflection from the integral: root by bisection, then Simpson
`series`	three-term weak-field series
`fig_potential`, `fig_paths`	figures

Run with `python bh_lensing.py` (needs NumPy and Matplotlib; about 30 seconds). It prints all tables above and saves five figures next to the file.

References

[1] V. Bozza, "Gravitational lensing in the strong field limit", Phys. Rev. D 66, 103001 (2002), arXiv:gr-qc/0208075.

[2] K. S. Virbhadra and G. F. R. Ellis, "Schwarzschild black hole lensing", Phys. Rev. D 62, 084003 (2000), arXiv:astro-ph/9904193.

[3] C. R. Keeton and A. O. Petters, "Formalism for testing theories of gravity using lensing by compact objects. I. Static, spherically symmetric black holes", arXiv:gr-qc/0511019 (weak-field series for the deflection angle).

[4] Event Horizon Telescope Collaboration, "First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole", Astrophys. J. Lett. 875, L1 (2019).

[5] S. Carroll, *Spacetime and Geometry: An Introduction to General Relativity*, Addison-Wesley (2004). (Schwarzschild geodesics and light deflection.)

[6] J. B. Hartle, *Gravity: An Introduction to Einstein's General Relativity*, Addison-Wesley (2003).